

Speculative Hardware Interfaces: Physical Modeling, Tactile Resonators, and Expressive Controllers

Author: Dr. Tamsin Callow • Co-authors: Prof. Dr. Julian Vance, Kaveh Danesh

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ABSTRACT

Computer music has historically separated the intangible flexibility of software from the physical tactile feedback of acoustic instruments. This paper presents our research into speculative hardware interfaces: performance controllers that combine physical computing, motorized haptic faders, microtonal touchplates, and tactile acoustic transducers. By vibrating physical woods, metals, and ceramics directly beneath the performer's fingers, our instruments reintroduce the intimate tactile resistance of traditional luthiery to algorithmic software synthesis.

KEY THEOREMS & BOUNDARY EQUATIONS

Formulation [1.1]

Tactile Feedback Coherence: Musicians achieve greater pitch accuracy and expressive nuance when software synthesis parameters produce corresponding physical vibrations in the tactile controller.

Formulation [2.1]

Haptic Resistance Coupling: Motorized faders programmed with programmable spring tension, detents, and friction can physically convey synthetic scale structures and algorithmic boundaries.

Formulation [3.1]

Material Resonance Synergy: Direct acoustic coupling of transducers to solid tone-woods creates an authentic acoustic dispersion field that bridges digital sound with natural room air.

1. The Tactile Disconnect in Digital Performance

Standard MIDI controllers and touchscreen glass offer visual feedback but are physically inert. An acoustic violinist senses string tension, fingerboard friction, and resonant wood vibrations through their fingertips and jawbone. This continuous haptic feedback loop enables subtle microtonal adjustments and intuitive expressive mastery.

2. Motorized Haptic Resistance Controllers

We integrated high-torque coreless DC motors onto custom linear faders, managed by a 32-bit ARM microcontroller running closed-loop PID control at 4 kHz. The fader can physically emulate elastic springs, viscous fluid resistance, ratcheted mechanical clicks for microtonal scales, or repulsive force fields corresponding to generative algorithmic boundaries.

3. Direct Material Transduction Surfaces

In our expressive keyboard prototypes, each key is carved from quarter-sawn spruce or sintered basalt and backed by a miniature voice-coil actuator. When an oscillator voice is triggered, the key vibrates with the exact waveform and overtone spectrum of the sound, giving the performer tangible physical feedback of envelope attack, filter sweep, and harmonic density.

4. Future Directions: Speculative Luthiery

We envision future musical instruments not as passive software controllers, but as symbiotic physical objects whose mechanical properties dynamically adapt to algorithmic composition, blurring the boundary between traditional acoustic craft and speculative technology.

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